# Questions

## Question 1: Predict the Output (Variables and Type Conversion)

Predict the output of the following Python program without executing it. Explain the data type and value of each variable.

```
x = 10
y = x
x = "20"
z = int(x)
result = str(y) + x
final_result = int(result) + z
print(x)
print(y)
print(z)
print(result)
print(final_result)
```

Also answer:

1. Why does the value of **y** remain unchanged when **x** is reassigned?

2. What would happen if **x = "20.5"** instead of **x = "20"**?

## Question 2: E-commerce Final Bill Calculator

An e-commerce company calculates the final bill using the following rules.

**Membership Discounts:**

| Membership | Discount |
|---|---|
| Gold | 20% |
| Silver | 10% |
| Regular | 0% |

**Additional Rules:**

1. Apply the membership discount first.

2. If the remaining amount is at least Rs. 5,000, apply an additional 5% discount.

3. If the payment method is online and the amount before the payment discount is at least Rs. 2,000, deduct another Rs. 200.

4. If the final amount is below Rs. 1,000, classify the order as Small Order.

5. If the final amount is between Rs. 1,000 and Rs. 5,000, inclusive, classify it as Standard Order.

6. If the final amount exceeds Rs. 5,000, classify it as Premium Order.

Write a Python program that accepts the purchase amount, membership type and payment method, then displays the total discount, final payable amount and order category.

## Question 3: Find the Hidden Logical Error

A college determines whether a student passes using these rules:

- Marks must be at least 40.

- Attendance must be at least 75%.

- If marks are at least 90, attendance is not required.

A student writes the following program:

```
marks = int(input("Enter marks: "))
attendance = float(input("Enter attendance: "))
if marks >= 40 and attendance >= 75 or marks >= 90:
    print("Pass")
else:
    print("Fail")
```

**Task:** Is the program logically correct? If not, identify the issue, correct the program, and demonstrate a test case that exposes the error.

## Question 4: Debug a Type-Conversion and Eligibility Program

A training institute offers admission if a student's marks are at least 60 and the student has paid a registration fee of at least Rs. 1,000. The following program is intended to implement the rule, but it contains mistakes.

```
marks = input("Enter marks: ")
fee = input("Enter registration fee: ")
if marks >= 60 and fee >= 1000:
    print("Admission Granted")
else:
    print("Admission Denied")
```

**Tasks:**

1. Identify why the program does not work as intended.

2. Correct the program without changing the admission rules.

3. Explain how the corrected program handles the inputs **60** and **1000**.

## Question 5: Variable Reassignment and Type Conversion

Predict the output.

```
a = 4
b = "3"
a = a + int(b)
b = str(a) + b
print(a)
print(b)
print(type(a))
print(type(b))
```